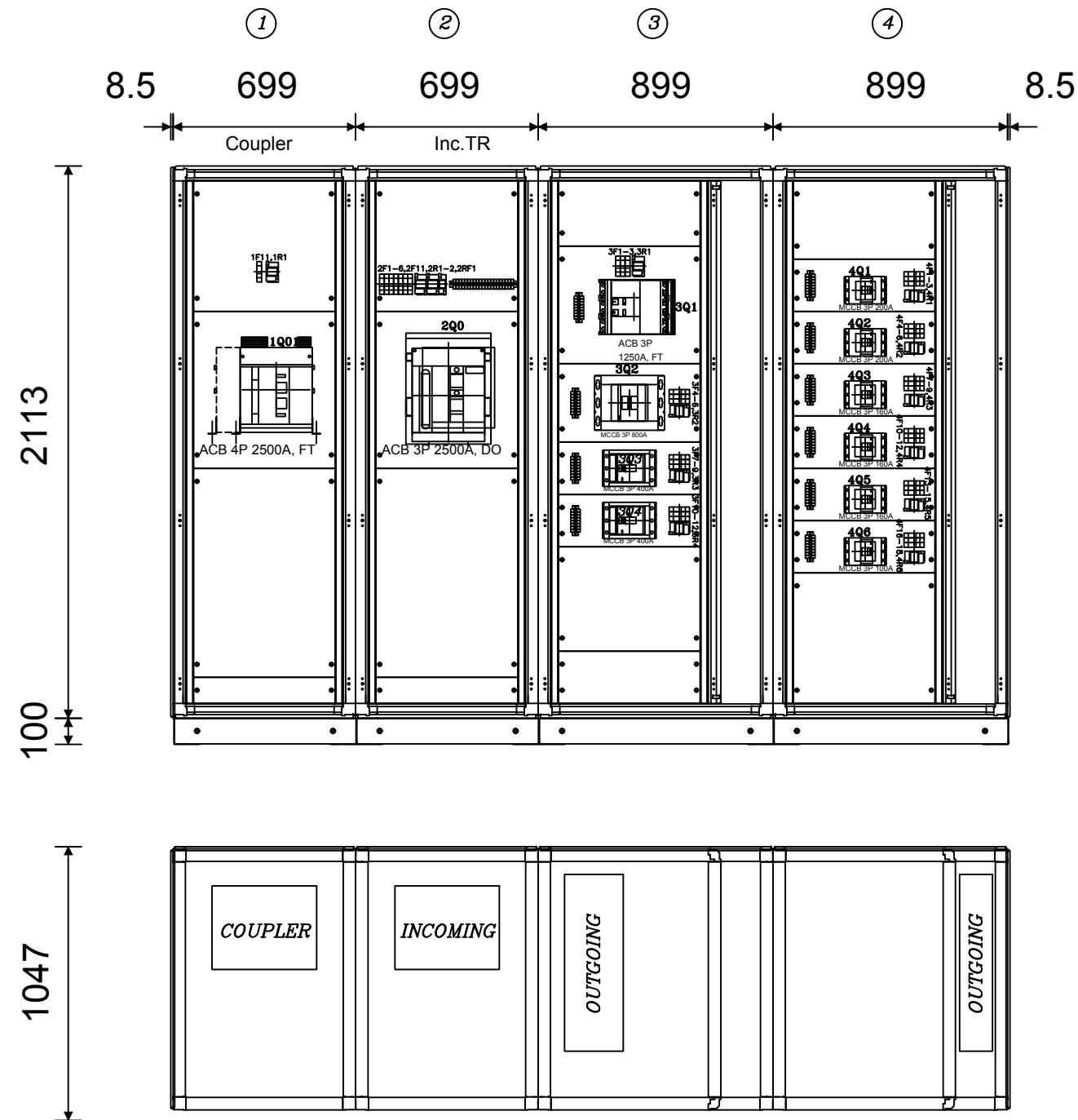
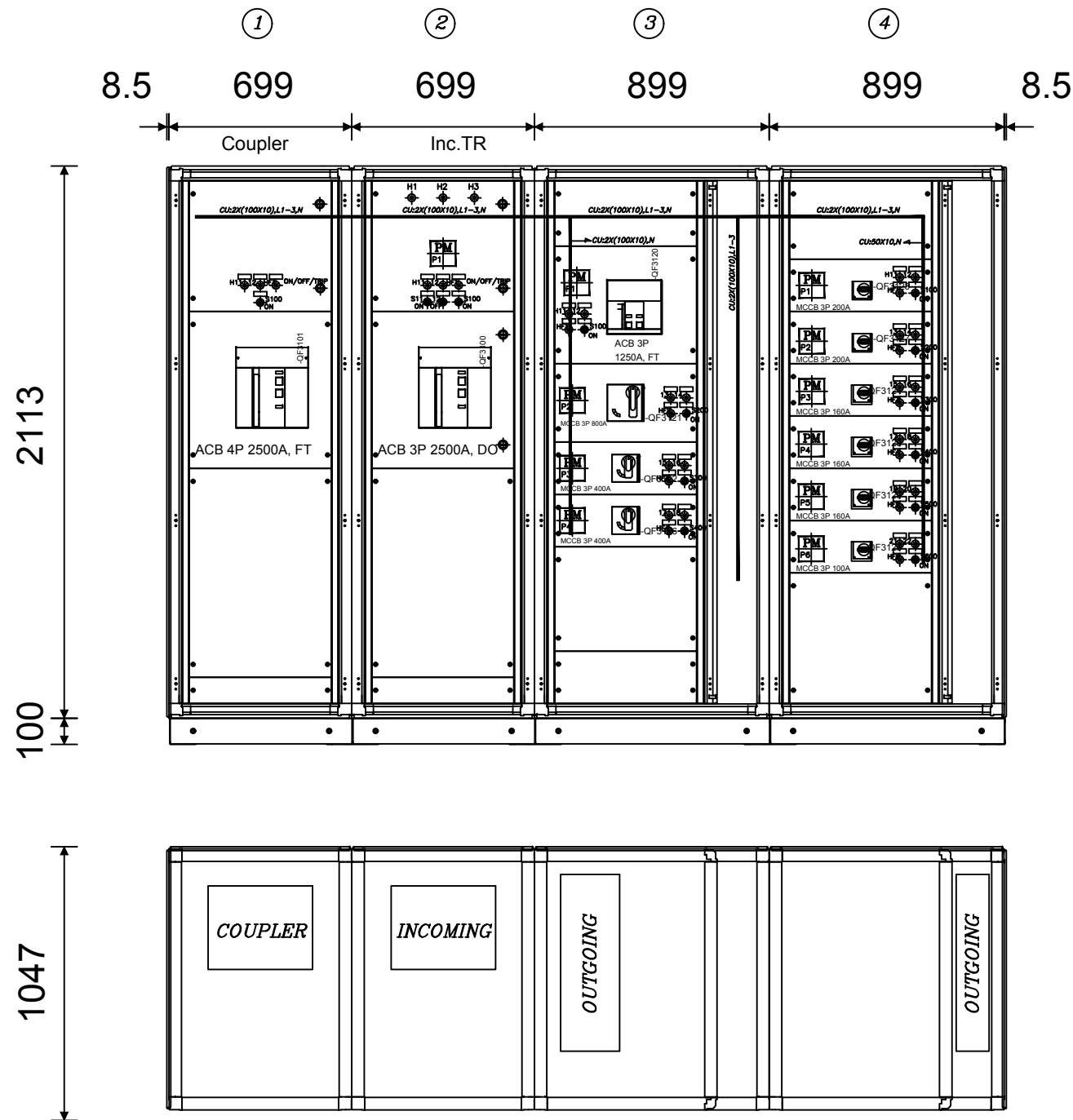


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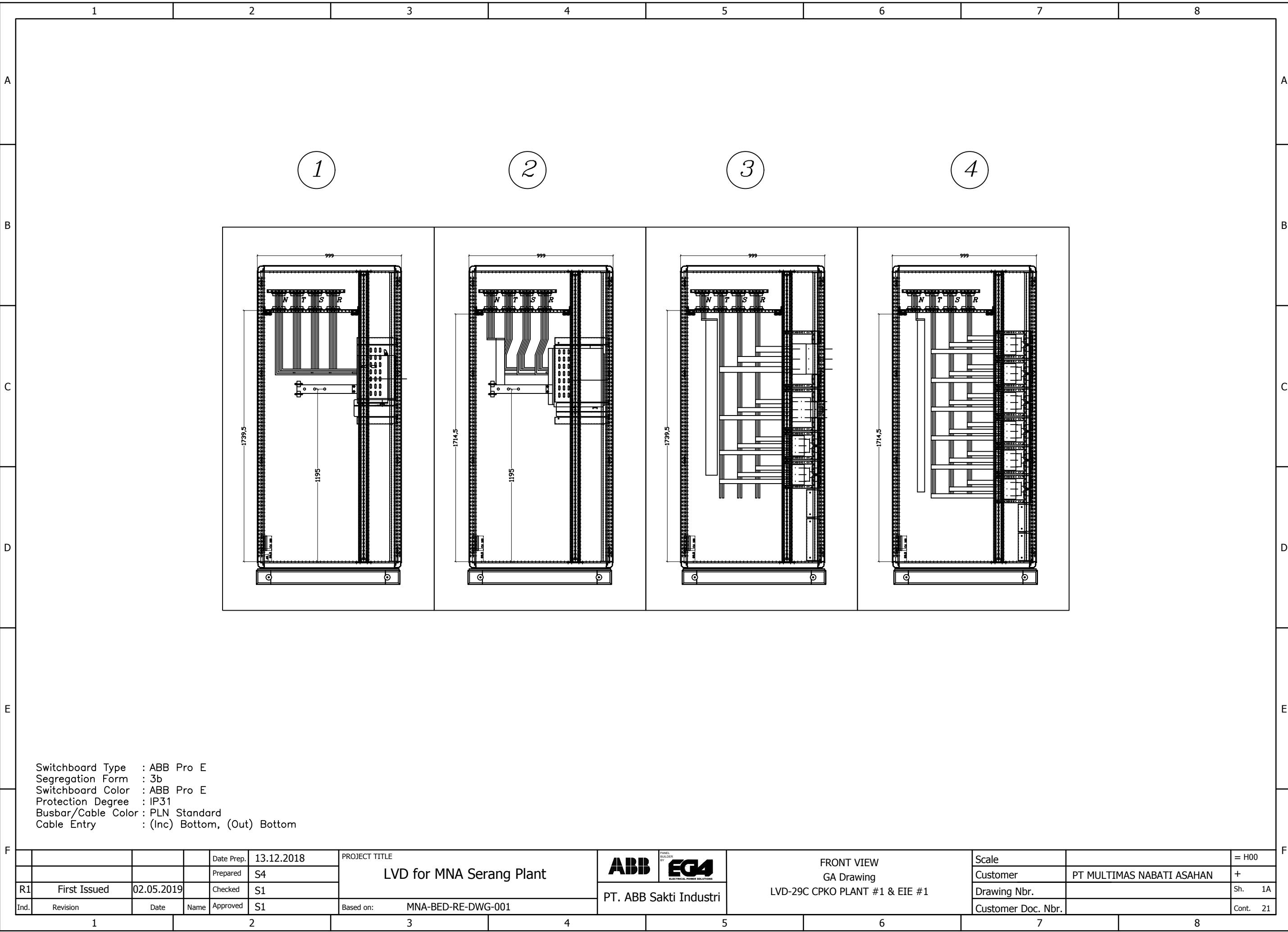
FRONT VIEW
LVD-29C CPKO PLANT #1 & EIE #1



Switchboard Type : ABB Pro E
Segregation Form : 3b
Switchboard Color : ABB Pro E
Protection Degree : IP31
Busbar/Cable Color : PLN Standard
Cable Entry : (Inc) Bottom, (Out) Bottom

R1		First Issued	02.05.2019	Checked	S1	PROJECT TITLE LVD for MNA Serang Plant Based on: MNA-BED-RE-DWG-001	ABB EG4 PT. ABB Sakti Industri	FRONT VIEW GA Drawing LVD-29C CPKO PLANT #1 & EIE #1	Scale	= H00
Ind.		Revision	Date	Name	Approved				Customer	PT MULTIMAS NABATI ASAHAN
									Drawing Nbr.	Sh. 1
									Customer Doc. Nbr.	Cont. 21

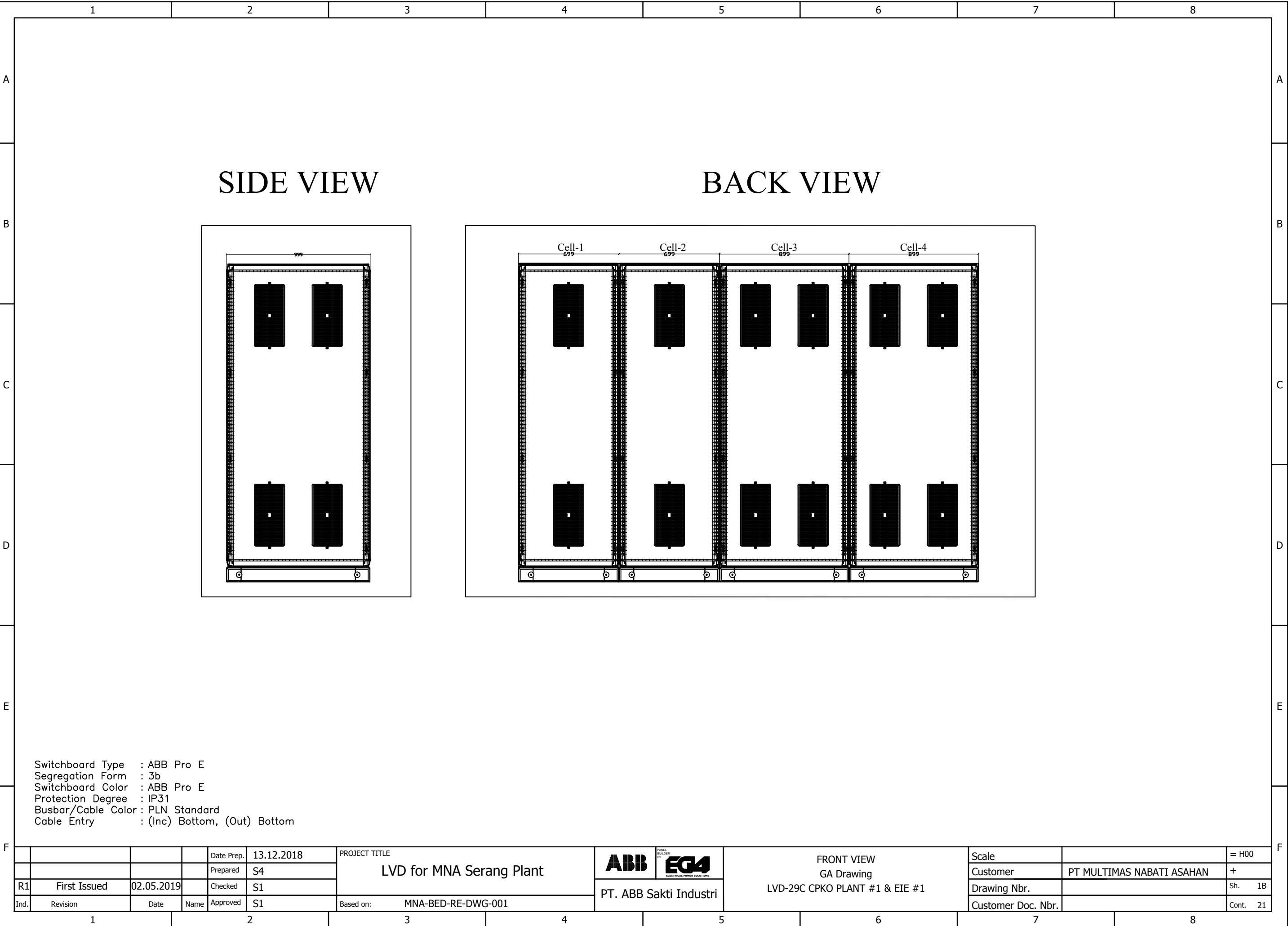
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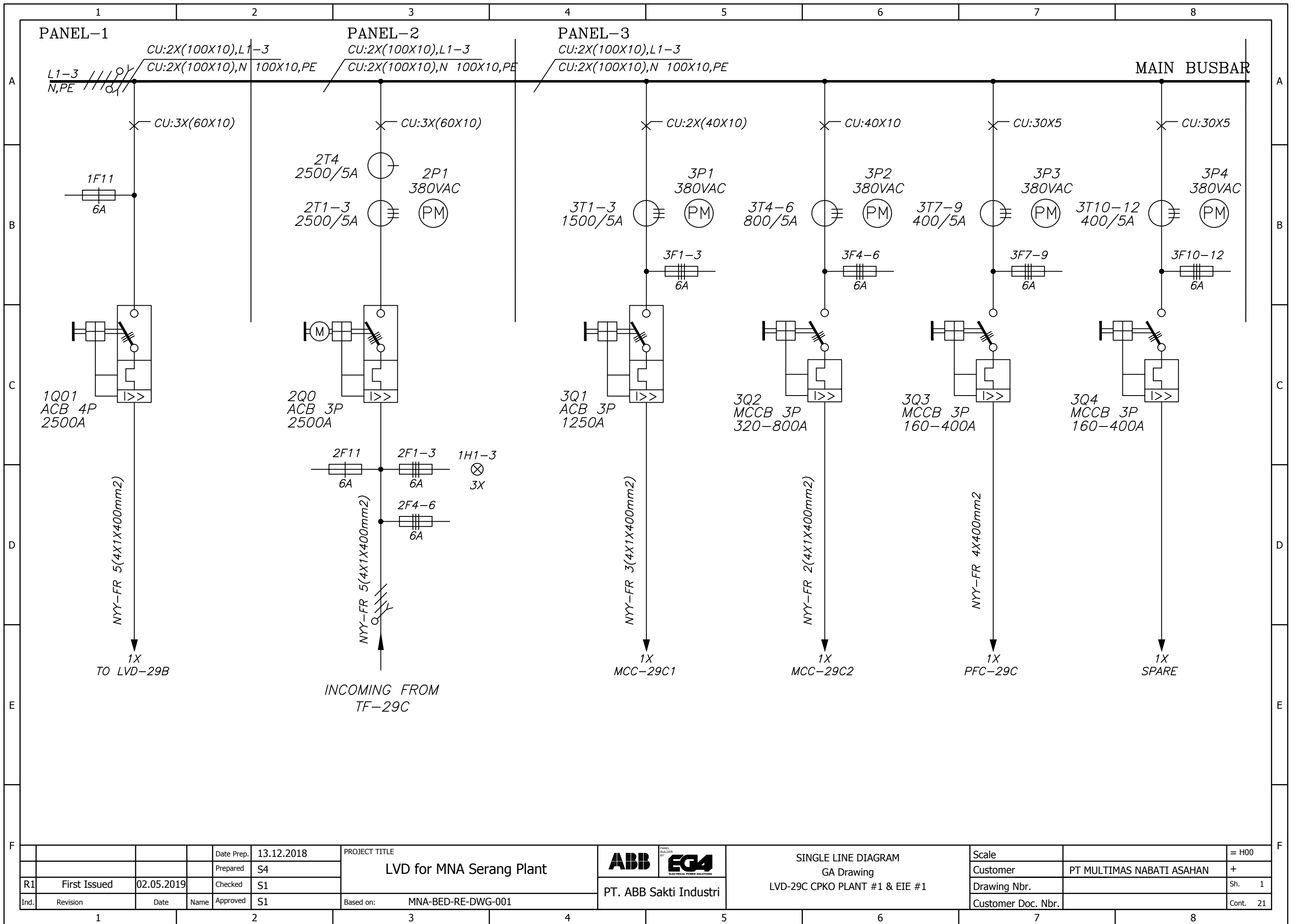
Switchboard Type : ABB Pro E
Segregation Form : 3b
Switchboard Color : ABB Pro E
Protection Degree : IP31
Busbar/Cable Color : PLN Standard
Cable Entry : (Inc) Bottom, (Out) Bottom

				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDING BY EG4 ELECTRICAL POWER SOLUTIONS PT. ABB Sakti Industri	FRONT VIEW GA Drawing LVD-29C CPKO PLANT #1 & EIE #1	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
R1	First Issued	02.05.2019		Checked	S1				Drawing Nbr.		Sh. 1A
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001		Customer Doc. Nbr.		Cont. 21

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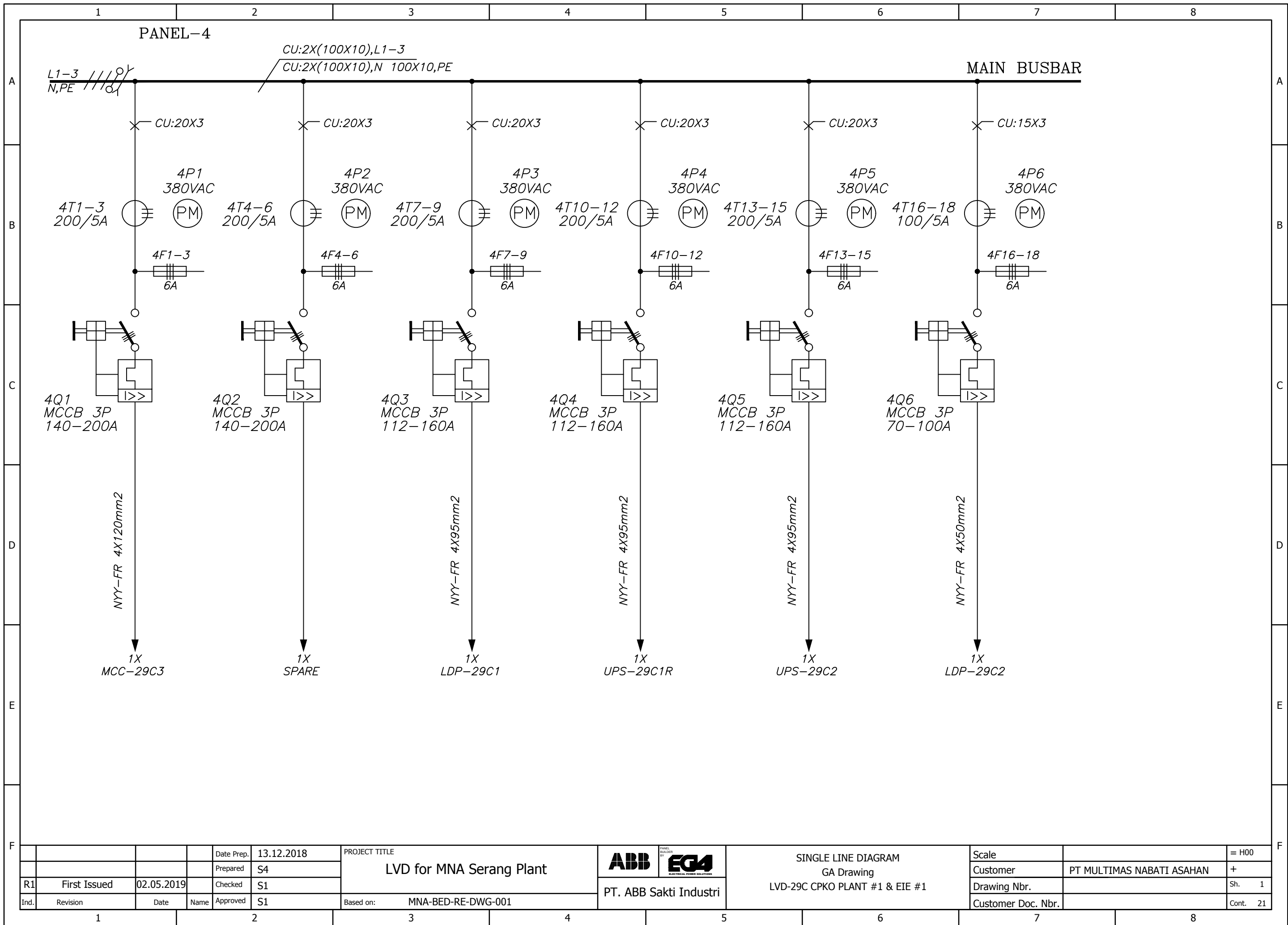


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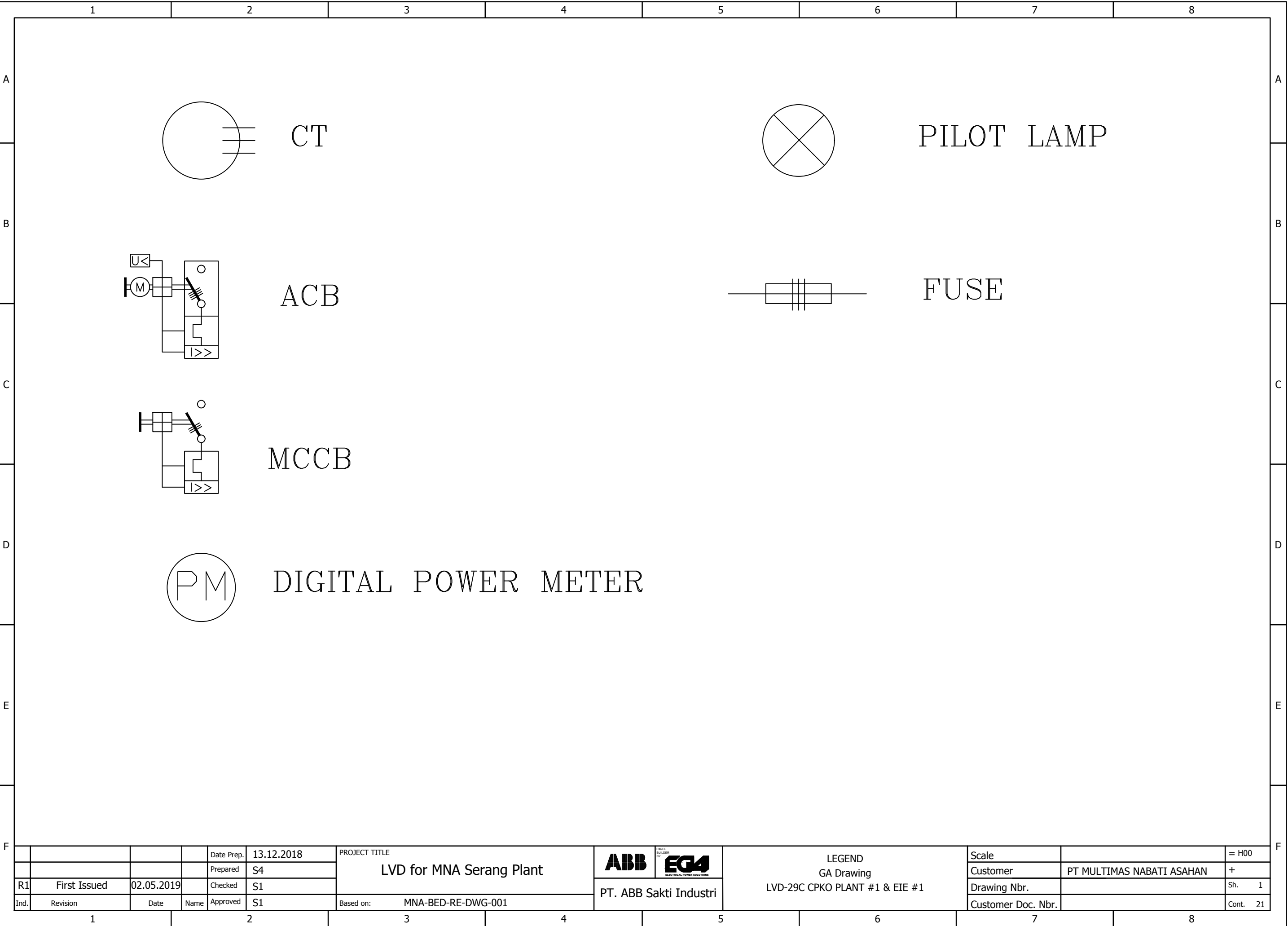


			Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDING BY EG4 ELECTRICAL POWER SOLUTIONS	SINGLE LINE DIAGRAM GA Drawing LVD-29C CPKO PLANT #1 & EIE #1	Scale		= H00
			Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
R1	First Issued	02.05.2019	Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001	PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 21

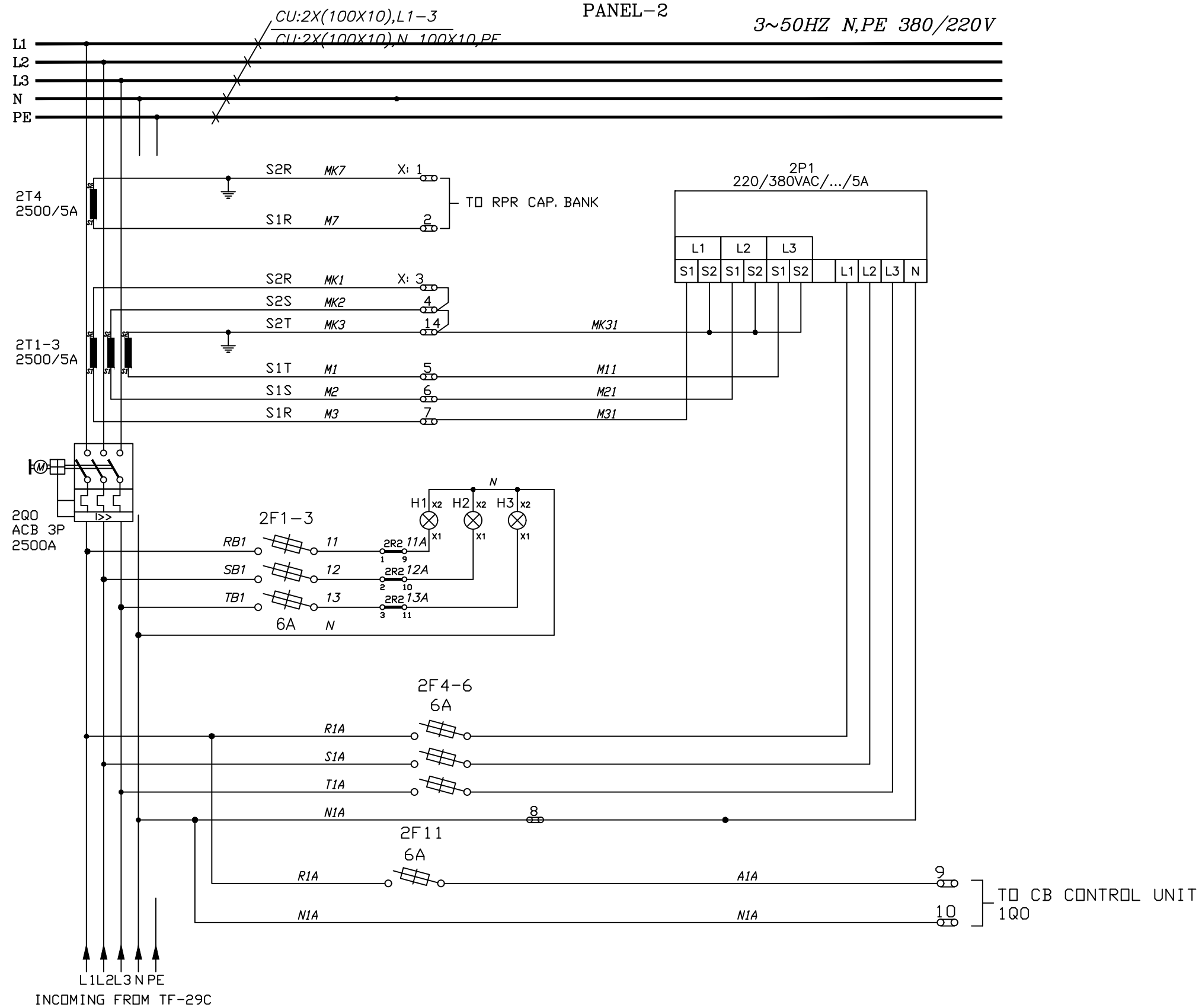
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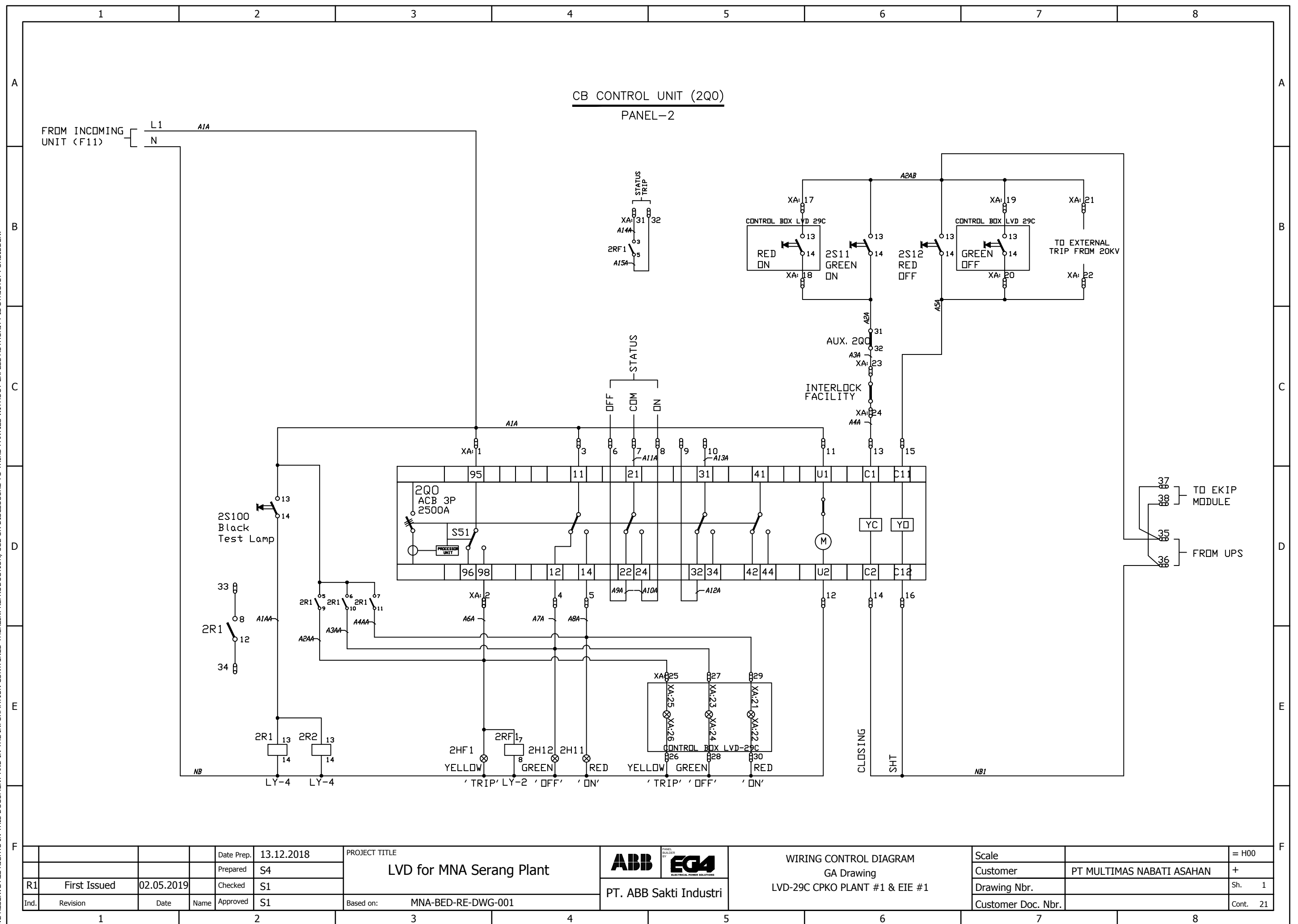


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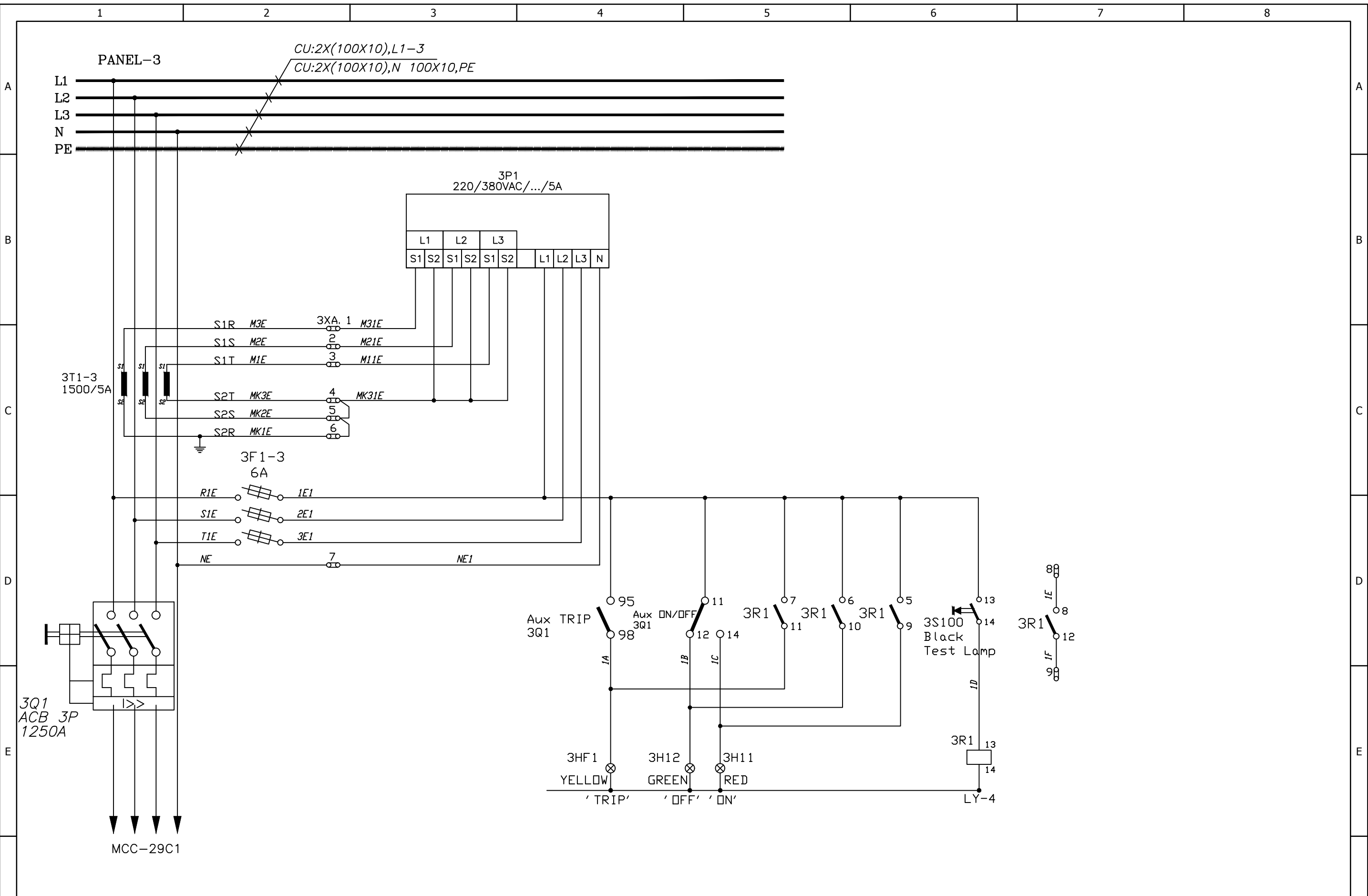


				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	  PT. ABB Sakti Industri	WIRING CONTROL DIAGRAM GA Drawing LVD-29C CPKO PLANT #1 & EIE #1	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
R1	First Issued	02.05.2019		Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001		Customer Doc. Nbr.		Cont. 21

1 2 3 4 5 6 7 8

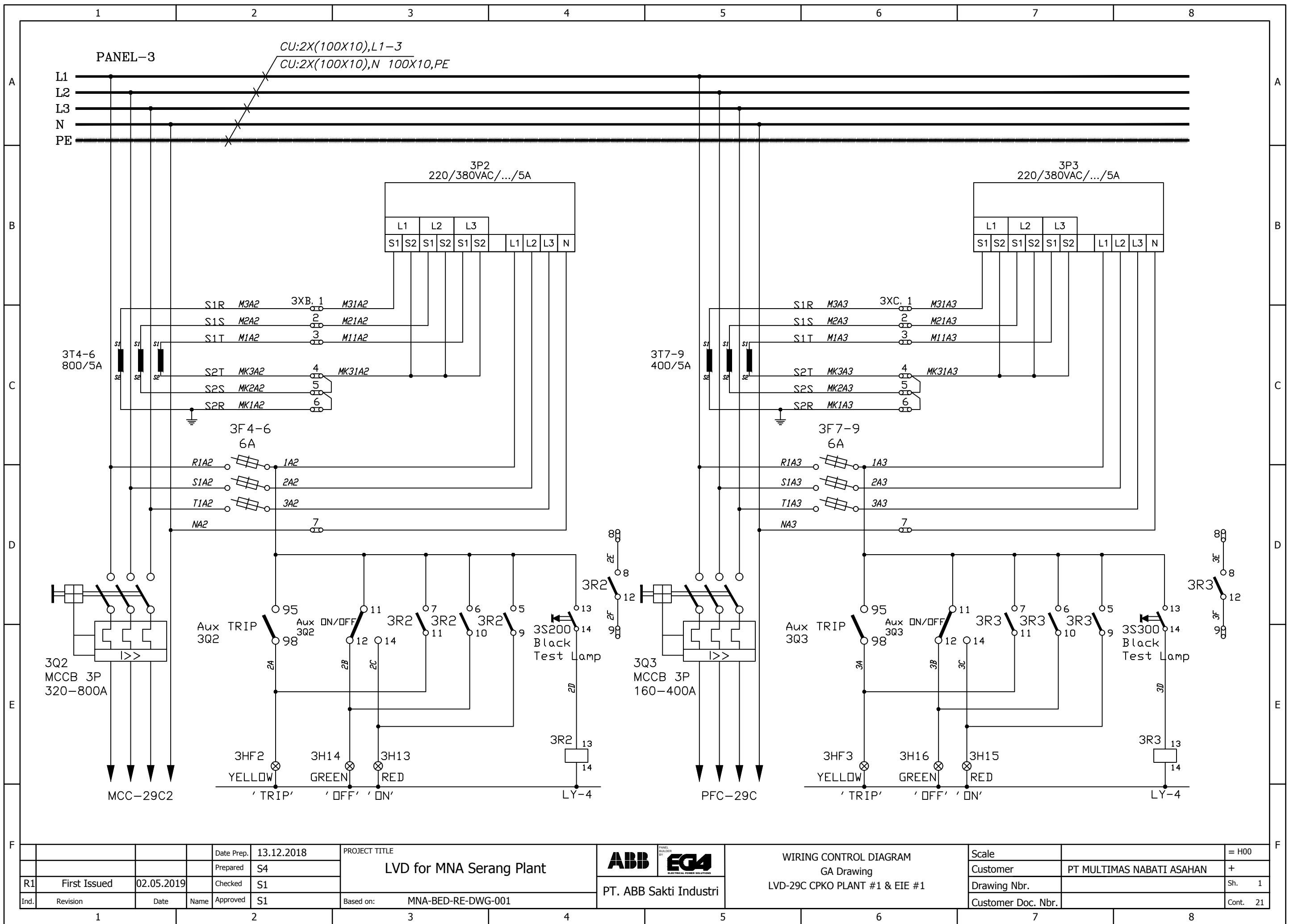


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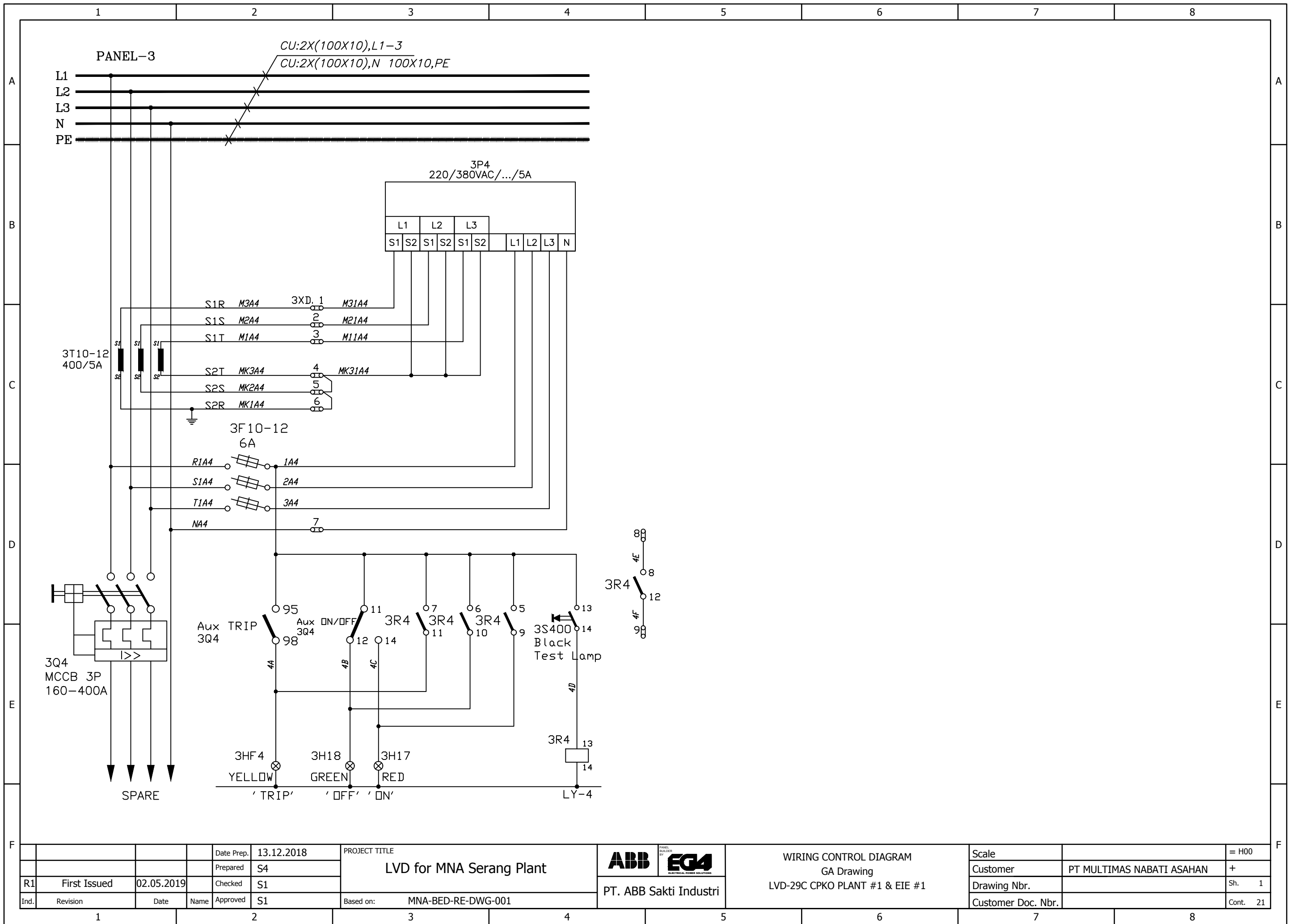


				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	 	WIRING CONTROL DIAGRAM GA Drawing LVD-29C CPKO PLANT #1 & EIE #1	Scale		= H00						
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+						
R1	First Issued	02.05.2019		Checked	S1				Drawing Nbr.		Sh. 1						
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 21						
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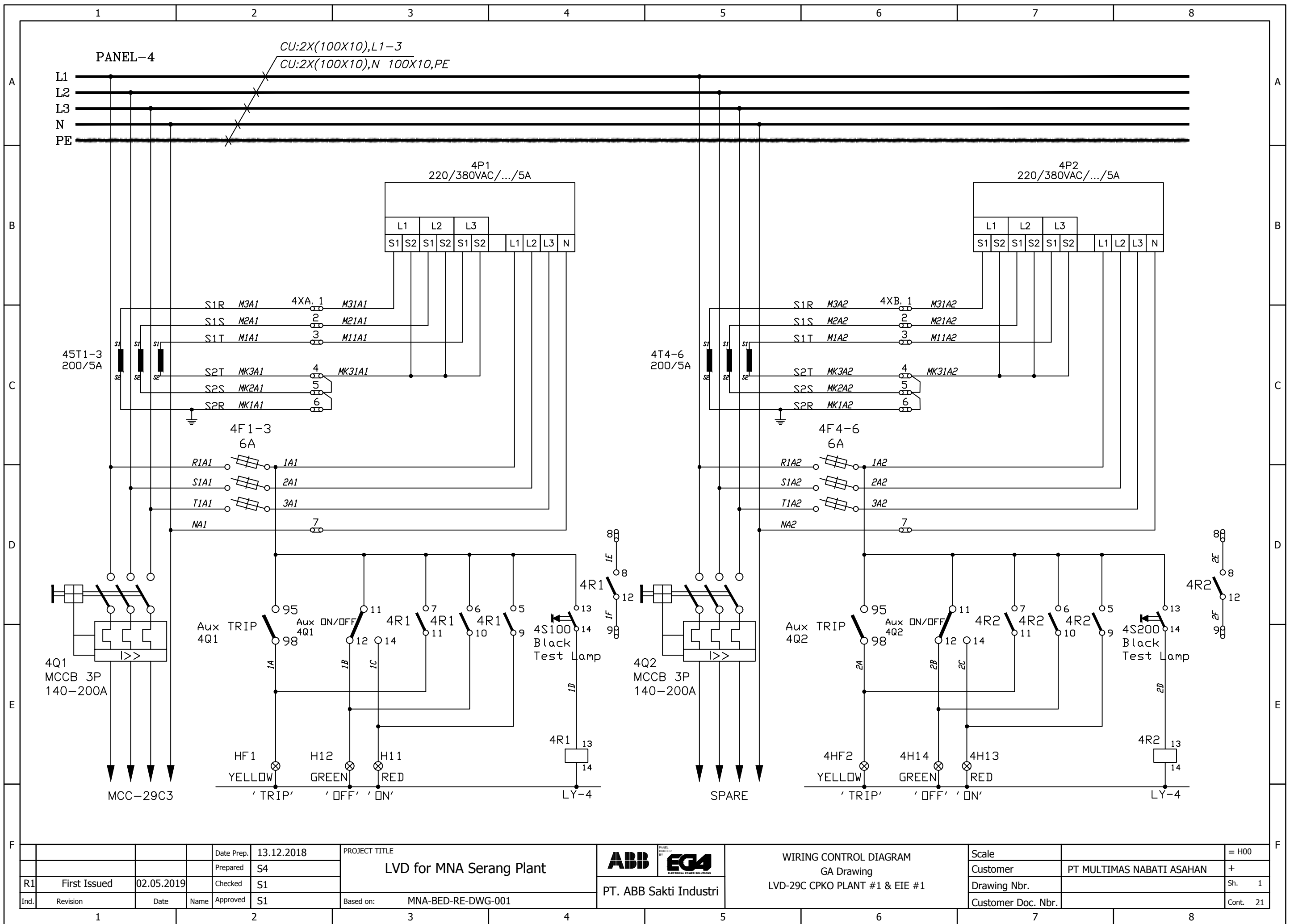


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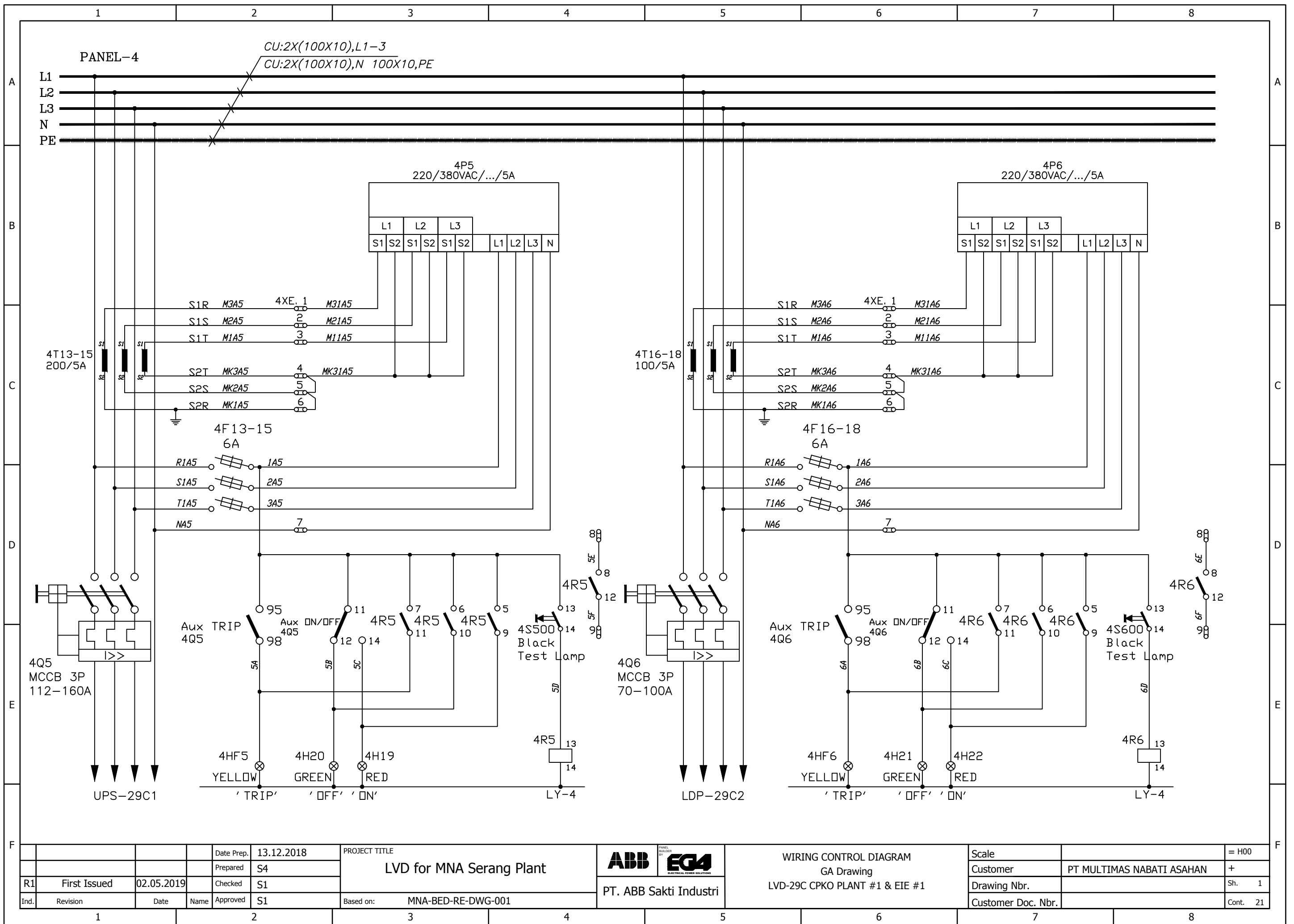


		Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		WIRING CONTROL DIAGRAM GA Drawing LVD-29C CPKO PLANT #1 & EIE #1		Scale	= H00	
		Prepared	S4					Customer	PT MULTIMAS NABATI ASAHAN	
R1		First Issued	02.05.2019	Checked	S1			Drawing Nbr.	Sh. 1	
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.	Cont. 21	

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			Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL DRAWING BY EG4 ELECTRICAL POWER SOLUTIONS	WIRING CONTROL DIAGRAM GA Drawing LVD-29C CPKO PLANT #1 & EIE #1	Scale		= H00
			Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
R1	First Issued	02.05.2019	Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001	PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 21

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A	N O	DESCRIPTION			TECHNICAL DATA			QTY	DESIGNATION						A				
		INCOMING																	
B	1	Fuse Carrier 6A			AXLE/Setara			7	2F11,2F1-6						B				
	2	Pilot Lamp (LED) 220Vac/50Hz, Red			AXLE/Setara			2	2H1,11										
	3	Pilot Lamp (LED) 220Vac/50Hz, Yellow			AXLE/Setara			2	2H2,HF1										
	4	Pilot Lamp (LED) 220Vac/50Hz, Green			AXLE/Setara			2	2H3,12										
	5	DIGITAL POWER METER (*)			ABB, M2M ETHERNET			1	2P1										
	6	CT 2500/5A CL.0.5 40VA			Howig, HG 120-2500			4	2T1-4										
	7	ACB 3P 2500A, 66KA, DO, Ekip Touch LSIG (*)			ABB, E2.2N25 D/O			1	2Q0										
C	8	+ Shunt Trip Release (Y0) (*)			ABB, Y0			1	-						C				
	9	+ Motor (M) + Closing Release (YC) For E2.2-E6.2 (*)			ABB, M+YC			1	-										
	10	+ Padlocks For E2.2-E6.2 (*)			ABB,			1	-										
	11	+ Ekip Com Protocol IEC 61850 (*)			ABB,			1	-										
	12	Push Button On, 1NO, Green			Ega			1	2S11										
	13	Push Button Off, 1NO, Red			Ega			1	2S12										
	14	Control Relay 220Vac			Omron, LY-2			1	2RF1										
D	15	Control Relay 220Vac			Omron, LY-4			2	2R1-2						D				
	16	Push Button Test Lamp , 1NO, Black			Ega			1	2S100										
	17																		
	18	COUPLER																	
	19	Fuse Carrier 6A			AXLE/Setara			1	1F11										
	20	Pilot Lamp (LED) 220Vac/50Hz, Red			AXLE/Setara			1	1H11										
	21	Pilot Lamp (LED) 220Vac/50Hz, Yellow			AXLE/Setara			1	1HF1										
E	22	Pilot Lamp (LED) 220Vac/50Hz, Green			AXLE/Setara			1	1H12						E				
	23	ACB 4P 2500A, 66KA, FT, Ekip Dip LI (*)			ABB, E2.2N25 F/T			1	1Q01										
	24	+ Padlocks For E2.2-E6.2 (*)			ABB,			1	-										
	25	Push Button Test Lamp , 1NO, Black			Ega			1	1S100										
	26	Control Relay 220Vac			Omron, LY-4			1	1R1										
	F					Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant			  PT. ABB Sakti Industri		COMPONENT LIST GA Drawing LVD-29C CPKO PLANT #1 & EIE #1			Scale			= H00
					Prepared	S4	Customer								PT MULTIMAS NABATI ASAHAN		+		
R1		First Issued	02.05.2019		Checked	S1	Drawing Nbr.										Sh.	1	
Ind.		Revision		Date	Name	Approved	S1	Customer Doc. Nbr.							Cont.	21			
1		2		3		4		5		6		7		8					

			Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	 	COMPONENT LIST GA Drawing LVD-29C CPKO PLANT #1 & EIE #1	Scale		= H00
			Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
R1	First Issued	02.05.2019	Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001	PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 21

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1		2		3		4		5		6		7		8	
A	N O		DESCRIPTION				TECHNICAL DATA		QTY	DESIGNATION					
			OUTGOING												
B	1	MCCB 3P 112-160A, 50KA (*)				ABB, XT2S TMA-160		3	4Q3-5						
	2	+ Auxiliary Contact For XT2/XT4 (*)				ABB, Aux-C 2Q 1SY		3	-						
	3	+ Extended Rotary Handles For XT2/XT4 (*)				ABB, RHE		3	-						
	4	MCCB 3P 70-100A, 50KA (*)				ABB, XT2S TMA-100		1	4Q6						
	5	+ Auxiliary Contact For XT2/XT4 (*)				ABB, Aux-C 2Q 1SY		1	-						
	6	+ Extended Rotary Handles For XT2/XT4 (*)				ABB, RHE		1	-						
C	7	Control Relay 220Vac				Omron, LY-4		10	3R1-4,4R1-6						
	8	Push Button Test Lamp , 1NO, Black				Ega		10	3S100-400 ;4S100-600						
	9														
	10														
	11														
	12														
	13														
	14														
	15														
	16														
D	17														
	18														
	19														
	20														
	21														
	22														
E	23														
	24														
	25														
	26														
F				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		 	COMPONENT LIST		Scale		= H00		
				Prepared	S4				GA Drawing		Customer	PT MULTIMAS NABATI ASAHAN	+		
	R1	First Issued	02.05.2019	Checked	S1				LVD-29C CPKO PLANT #1 & EIE #1		Drawing Nbr.		Sh. 1		
	Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 21			
	1		2		3		4		5		6		7		8